

## Unit 9, Lesson 4: Writing a Quadratic Function From a Written Description



### Benchmarks

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.



### Learning Targets

- ☐ I can write a quadratic function to represent the relationship between two quantities from a written description.
- ☐ I can rewrite a quadratic equation from vertex form to standard form.
- ☐ I can solve for a variable of interest.



### 9.4.1 Warm-Up: Words, Tables, and Graphs - Part 2

In the prior Warm-Up, the graphs were matched to a table of values.

1. Match the same graphs to the appropriate written description.

| Graphs   | Written Description                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| I.<br>   | A.<br>The weight of your jumbo box of cereal decreases by an equal amount every week.                                 |
| II.<br>  | B.<br>The tennis ball is dropped off the roof of a skyscraper.                                                        |
| III.<br> | C.<br>For a while it looked like the decline in profits was slowing down, but then they began declining more rapidly. |

2. Discuss with your partner the strategy you used to match the verbal descriptions and graphs.



### 9.4.2 Collaboration: Airline Tickets

An airline has a new plane with 120 seats and a new route from Tampa, FL to Cleveland, OH. The airline will maximize profit for this route at \$2,358 per flight. Based on a comprehensive customer survey, the airline determined that charging \$55 a seat will return a profit of \$2,100. They also found that charging \$75 a seat will return a profit of \$2,100. After analyzing all of the data, the airline determined that charging \$65 a seat would allow the company to maximize the profit.

- Complete the statements.

The maximum profit for the route is ☐ \$2,100. ☐ \$2,358. The

seat charge for customers when the airline has

achieved the maximum profit is ☐ \$55. ☐ \$65. ☐ \$75.

The point at which the airline has maximized their profit is the vertex  $(x, f(x))$ . The  $x$ -value of the vertex is the seat charge at the point when the company has a maximum profit, where the maximum profit is the  $f(x)$  value of the vertex.

- The vertex in the context is ☐ (55, 2100) ☐ (65, 2358) ☐ (75, 2100) because

this is the point where the ☐ customers will pay the most per seat. ☐ airline will make the most money.

The profit the airline makes is a function of the cost per seat. Because the profit has a maximum and two points that are equidistant from the vertex, a quadratic function can be used to model the context. In either standard form as  $f(x) = ax^2 + bx + c$  or in vertex form  $f(x) = a(x - h)^2 + k$ , where the variable  $x$  is the cost per seat and  $f$  is the profit.

- Discuss with your partner how to confirm that the two points (55, 2100) and (75, 2100) are equidistant from the vertex.

- Use the steps in the table to write the quadratic function that models the airline's profit.

|                                               |                         |
|-----------------------------------------------|-------------------------|
| Substitute values for the point $(x, f(x))$ . | $f(x) = a(x - h)^2 + k$ |
| Substitute values for the vertex $(h, k)$ .   | $f(x) = a(x - h)^2 + k$ |
| Solve for the variable $a$ .                  | $f(x) = a(x - h)^2 + k$ |

Write the equation for the quadratic function in vertex form.

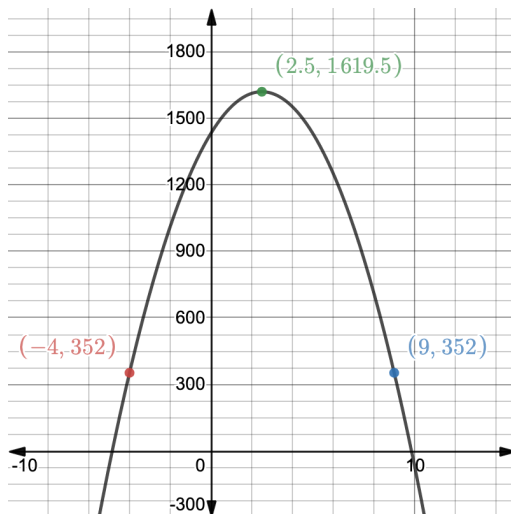


### 9.4.3 Try It: Writing Quadratic Functions

Work with your partner on the following.

1. Frieda is creating a quadratic equation to represent the number of movie tickets sold in the U.S., in millions, from 2000 to 2005. She determines the number of tickets sold,  $T$ , is a function of the year.

| $x$  | $T(x)$ |
|------|--------|
| -4   | 352    |
| -3.5 | 539.5  |
| 2.5  | 1619.5 |
| 4.5  | 1499.5 |
| 9    | 352    |



Write the quadratic function in both vertex and standard form that models the context. Show your work. Then, write the equations in the table in problem 3.

2. A botanist was monitoring the growth of a type of tree during its first year. The botanist measured the heights of the trees in micrometers and the amount of rain they received in inches.

The botanist discovered the trees grew to an average height of 90,671 micrometers when there was 1 inch of rain or 43 inches of rain. The trees averaged the tallest height of 245,462 micrometers when there was 22 inches of rain.

Write the equation for the quadratic function of the average height of this type of tree in micrometers,  $H$ , as a function of the inches of rain,  $x$ , in vertex and standard form. Show your work below. Then, write the equations in the table in problem 3.

3. Ask another pair of students to check your work for the two quadratic equations you wrote for each context, while you and your partner check their work. If necessary, correct any errors. Each person should write their initials next to your equations stating that your equations are correct or were corrected after fixing any errors.

| Context       | Vertex Form | Standard Form | Initials |
|---------------|-------------|---------------|----------|
| Movie Tickets |             |               |          |
| Tree Height   |             |               |          |



### 9.4.4 Your Turn!

- Izaak and Mylee just bought a house with open land in the backyard. The previous owner left 400 linear feet of fencing. Izaak and Mylee want to construct the fence to give their two dogs the most area possible to play in the backyard. If they make the length of the fence 80 linear feet or 120 linear feet, the dogs will have a play area of 9,600 square feet. However, if Izaak and Mylee make the length of the fence 100 linear feet, they will be able to create a maximum area of 10,000 square feet.

The area of play for the dogs,  $A$ , in square feet is a function of the length of the fence,  $x$ , in linear feet.

- What is the point  $(x, y)$  where the dogs will have the maximum fenced in area?

- Identify one other point from the context.

| Vertex $(h, k)$ | Point $(x, A(x))$ |
|-----------------|-------------------|
| (____, ____)    | (____, ____)      |

- Using the vertex and point, write the quadratic function in vertex form.

**Vertex Form**

- An Olympic diver is preparing for his upcoming diving competition. He must execute his dive off of a 10 foot high diving board. In order to have enough time to execute all the twists and flips, after 2 seconds he needs to be at a height of 14 feet. After 4 seconds, he needs to be back at 10 feet above the water for his last flip as he finishes his dive.

The diver's height in feet,  $H$ , is a function of time in seconds,  $x$ .

- Identify the vertex and one additional point of the function from the context.

| Vertex $(h, k)$ | Point $(x, H(x))$ |
|-----------------|-------------------|
| (____, ____)    | (____, ____)      |

- Write the quadratic function in standard form.



### 9.4.5 Wrap-Up: Cool Down

The last three lessons have involved writing quadratic functions to represent the relationship between two quantities from a graph, a table, and a written description.

1. Create a sales pitch to convince your classmates which representation you feel is the easiest for writing quadratic functions.

## Unit 9, Lesson 5: Solving Quadratics Using Square Roots



### Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



### Learning Targets

- ☐ I can solve a quadratic by taking the square root.
- ☐ I can determine if a solution to a quadratic is non-real.
- ☐ I can interpret solutions to quadratics given in a real-world context as viable or nonviable.